Type MS · Substitute into a given equation, rearranging and converting units

Answer 1 of 9 · Question 3 part b.iv · 3 marks

2023 JUN · Paper 02 (regional R) · find part b.iv in this mark scheme extract

3 (a)	• contains particular features / characteristics/ adaptations; • to perform a specific function;		2
(b) (i)	• 84 ÷ 0.006; • 14000x	Allow correct final answer from a calculation that includes 84 and 0.006, max 1 mark	2
(ii)	half the amount/haploid/23 chromosomes;		1
(iii)	• to provide energy; • for swimming/movement/travelling;		2
(iv)	• 180 ÷ 0.070; • 2571; • (÷ 60) 42.85;	Allow any figure that rounds to 43 Allow ecf for third mark - an incorrect value for a calculation carried out for mps1 and 2, divided by 60 **and** a correct answer to that calculation	3 exp
(c)	Any two from: • greater magnification; • higher resolution/can see objects more clearly/in more detail; • can see a greater number of structures;		2 exp

Total Question 3 = 12 marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 2 of 9 · Question 5 part b.ii · 3 marks

2020 JAN · Paper 01 · find part b.ii in this mark scheme extract

Question number					
5	(a)	(i)	12 x 50 = 600 (1) 600 ÷ 1000 = 0.6 (1)	Allow 600 without working out. Allow full marks for correct final answer.	2
		(ii)	provides insulation/prevents nerve impulse dispersing/speeds up nerve impulse (1)		1
	(b)	(i)	• hormones are transported in blood (1) • speed of blood flow is determined by heart rate/hormones can only travel at the speed blood is moving/blood flow slower (1)	Allow reverse argument for both marking points	2
		(ii)	• recognition of distance divided by time (1) • $1.0 \div 1.2 \times 10^{2}$ / $1.0 \times 10^{6} \div 1.2 \times 10^{8}$ (1) • 0.0083(3) / 8.3×10^{-3} (s) (1)	Allow full marks for correct final answer	3

Total for Question 5 = 8 marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 3 of 9 · Question 3 part c · 2 marks

2022 JUN · Paper 02 · find part c in this mark scheme extract

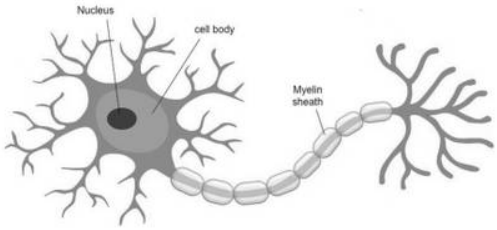
3 (a)	In the following order: • receptors; • sensory; • relay; • motor;		4
(b) (i)	synapse;		1
(ii)	• chemicals/neurotransmitters; • diffuse across synaptic clef/gap;		2
(c)	• 60 × 5; • 300 x 120 = 36 000;		2
			Total 9

Question number	Answer	Notes	Marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 4 of 9 · Question 7 part c · 4 marks

2024 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

7 (a) (i)	receptor (cell)		(1)
(ii)	sensory neurone		(1)
(b) (i)	a cell that has particular features/adaptations/structures; to carry out a certain/specific function in the body;		(2)
(ii)	diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending; nucleus; cell body; myelin sheath;  	Label marks can be awarded if they draw a sensory or relay neurone	(4)
(iii)	transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone; to effecter organs/muscle/gland;		(2)
(c)	100/4.0 x 10^{6}; = 0.000025 (s); = 0.025 (ms); = 2.5 x 10^{-2};	ecf for mps 2,3 and 4 Allow full marks for correct final answer	(4)

Total for Question 7 = 14 marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 5 of 9 · Question 1 part b · 3 marks

2025 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

1 (a)	• atria; • lungs; • vena cava; • ventricles; • body; • muscle; • systolic; • 120;		8
(b)	6.1 ÷ 68; = 0.0897 dm^3; = 89.7 (cm^3);	ACCEPT 90 (cm^3) Allow 1 mark for incorrect dm^3 in MP2 to correct conversion to cm^3	3
			Total = 11

Question number	Answer	Notes	Marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 6 of 9 · Question 1 part b.ii · 2 marks

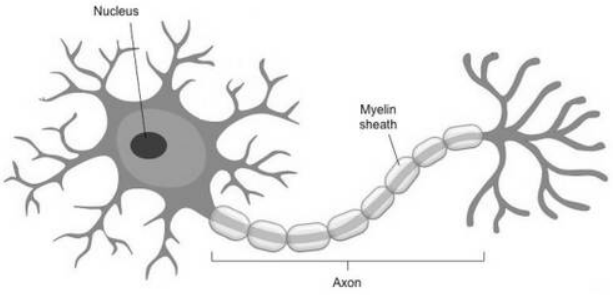
2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

1 (a)	(i)	long (bone)		1
	(ii)	<table><tr><th rowspan="2">Part</th><th colspan="3">Function</th></tr><tr><th>reduces friction</th><th>produces red blood cells</th><th>provides strength</th></tr><tr><td>compact bone</td><td></td><td></td><td>✓</td></tr><tr><td>spongy bone</td><td></td><td>✓</td><td></td></tr><tr><td>cartilage</td><td>✓</td><td></td><td></td></tr></table>		3
	(iii)	• provides nutrients/named nutrient/oxygen; • removes waste/named waste		2
(b)	(i)	In the following order: • tendons; • contracts; • relaxes; • pulls;		4
	(ii)	0.5 x 125; 62.5 (J);	Accept 63 (J)	2
			Total for Question 1 = 12 marks	

Type MS · Substitute into a given equation, rearranging and converting units

Answer 7 of 9 · Question 3 part b.iv · 2 marks

2025 JUN · Paper 02 (regional R) · find part b.iv in this mark scheme extract

3	(a)	(i)		1 mark for diagram 3 marks for labels	4
		(ii)	• transmits electrical/nerve impulses; • from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ;		2
	(b)	(i)	Any six from 1 (student X) holds up the ruler between the thumb and forefinger (of other student); 2 then drops/releases the ruler; 3 (other) student catches the ruler; 4 record the reading on the ruler/distance fallen when caught; 5 repeat for <u>this</u> student; 6 using the same distance between thumb and forefinger; 7 find mean/average value of reading on ruler; 8 calculate reaction time (using formula); 9 repeat with <u>other</u> students;		6
		(ii)	the reading on the ruler/ reaction time;		1
		(iii)	independent variable not continuous/is discrete;		1
		(iv)	reaction time = √(0.15 ÷ 5); reaction time = 0.17 (s);	accept 0.173(2)	2
				Total for Question 3 = 16 marks	

Type MS · Substitute into a given equation, rearranging and converting units

Answer 8 of 9 · Question 2 part b.i · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
2	(a) (i)	better transfer of heat to water (1)		1
	(ii)	any two from • safety goggles/glasses (1) • water may spit (1) • use test tube holder (1) • to avoid burning hands (1)	reason linked to precaution	2
	(b) (i)	• 28 x 12 x 4.2 (1) • 1411/1411.2J (1) • 1400J (1)	error carried forward full marks for correct answer	3
	(ii)	• some energy lost to air/absorbed by tube/needle (1) • some converted to light (1) • food not fully burnt;		3
	(c)	• use same mass of food (1) • same volume of water (1) • use same thin tube (1) • burn food same distance from tube (1)		4

Total 13 marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 8 of 9 · Question 2 part b.i · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 9 of 9 · Question 7 part b.i · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
7	(a)	$\frac{14}{100}$ x 250 = 35;	Ecf one mark for correct answer from wrong numbers	1
		35 ÷ 10 = 3.5	Full marks for correct final answer of 3.5	1
	(b) (i)	4 x 3.5 = 14 (units);	ecf from 7a	1
		14 – 3 = 11 (hours);	Full marks for correct final answer of 11 hours	1
	(ii)	Any two from: • alcohol is a depressant; • (alcohol) affects the nervous system/brain; • slows reactions/increases reaction <u>time</u>;		Max 2
	(c)	Any five from: • less/no bile produced; • less/no emulsification of fat; • surface area of fat remains low; • less fat digested/broken down (by lipase); • pH of small intestine remains low/acidic; • enzymes denatured/reduced enzyme activity; • reduced/no digestion by protease / carbohydrase/fewer proteins/carbohydrates digested/broken down; • more fat egested;	Allow trypsin for protease and amylase for carbohydrase	Max 5

Total question 7 = 11 marks

Type MS · Substitute into a given equation, rearranging and converting units

Answer 9 of 9 · Question 7 part b.i · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks